METABOLIC DISORDER OF AMINO ACIDS

INTRODUCTION:

- Twenty amino acids, including nine that cannot be synthesized in humans and must be obtained through food, are involved in metabolism.
- Amino acids are the building blocks of proteins; some also function as (or) are synthesized into important molecules in the body such as neurotransmitters, hormones, pigments and oxygen-carrying molecules.
- Amino acid metabolism disorders are **hereditary metabolic disorders**.
- Hereditary disorders occur when parents pass the defective genes that cause these disorders on to their children.
- Amino acids are the building blocks of proteins and have many functions in the body. Hereditary disorders of amino acid processing (metabolism) can result from defects either in the breakdown of amino acids or in the body's ability to get amino acids into cells.
- Disorders that affect the metabolism of amino acids include:
    1) Phenylketonuria
    2) Tyrosinemia
    3) Homocystinuria
    4) Non-ketotic hyperglycinemia and
    5) Maple syrup urine disease.
- These disorders are autosomal recessive, and all may be diagnosed by analyzing amino acid concentrations in body fluids.

1) PHENYLKETONURIA:

- Phenylketonuria (PKU) is caused by decreased activity of phenylalanine hydroxylase (PAH), an enzyme that converts the amino acid phenylalanine to tyrosine, a precursor of several important hormones and skin, hair, and eye pigments.
- Decreased PAH activity results in accumulation of phenylalanine and a decreased amount of tyrosine and other metabolites.
- Persistent high levels of phenylalanine in the blood in turn result in progressive developmental delay, a small head circumference, behaviour disturbances, and seizures.

- Due to a decreased amount of the pigment melanin, persons with PKLI tend to have lighter features, such as blond hair and blue eyes, than other family members who do not have the disease.
- Treatment with special formulas and with foods low in phenylalanine and protein can reduce phenylalanine levels to normal and maintain normal intelligence.
- However, rare cases of PKU that result from impaired metabolism of biopterin, an essential cofactor in the phenylalanine hydroxylase reaction, may not consistently respond to therapy.

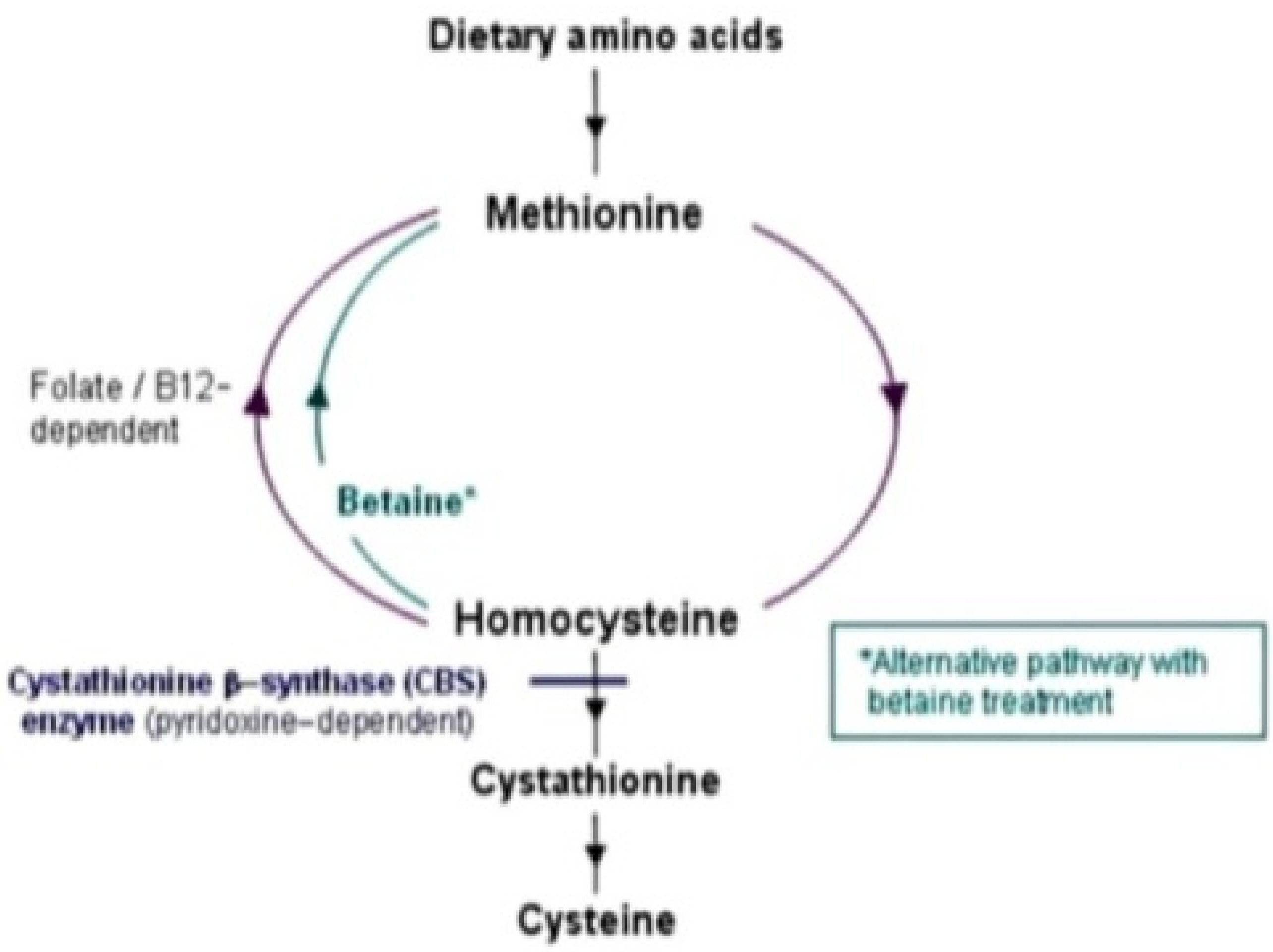
2) TYROSINEMIA:

- Classic (hepatorenal or type I) tyrosinemia is caused by a deficiency of fumarylacetoacetate hydrolase (FAH), the last enzyme in tyrosine catabolism.
- Features of classic tyrosinemia include severe liver disease, unsatisfactory weight gain, peripheral nerve disease, and kidney defects.
- Approximately 40 percent of persons with the disorder develop liver cancer by the age of 5 if untreated.
- Tyrosinemia or tyrosinaemia is an error of metabolism, usually inborn, in which the body cannot effectively break down the amino acid tyrosine.
- Symptoms of untreated tyrosinemia include liver and kidney disturbances.
- Treatment with 2-(2-nitro-4-trifluoromethylbenzoyI)-L3-cyclohexanedione (NTH), a potent inhibitor of the tyrosine catabolic pathway, prevents the production of toxic metabolites.
- Although this leads to improvement of liver, kidney, and neurological symptoms, the occurrence of liver cancer may not be prevented.
- Liver transplantation may be required for severe liver disease or if cancer develops.
- A benign, transient neonatal form of tyrosinemia, responsive to protein restriction and vitamin C therapy, also exists.

3) HOMOCYSTINURIA:

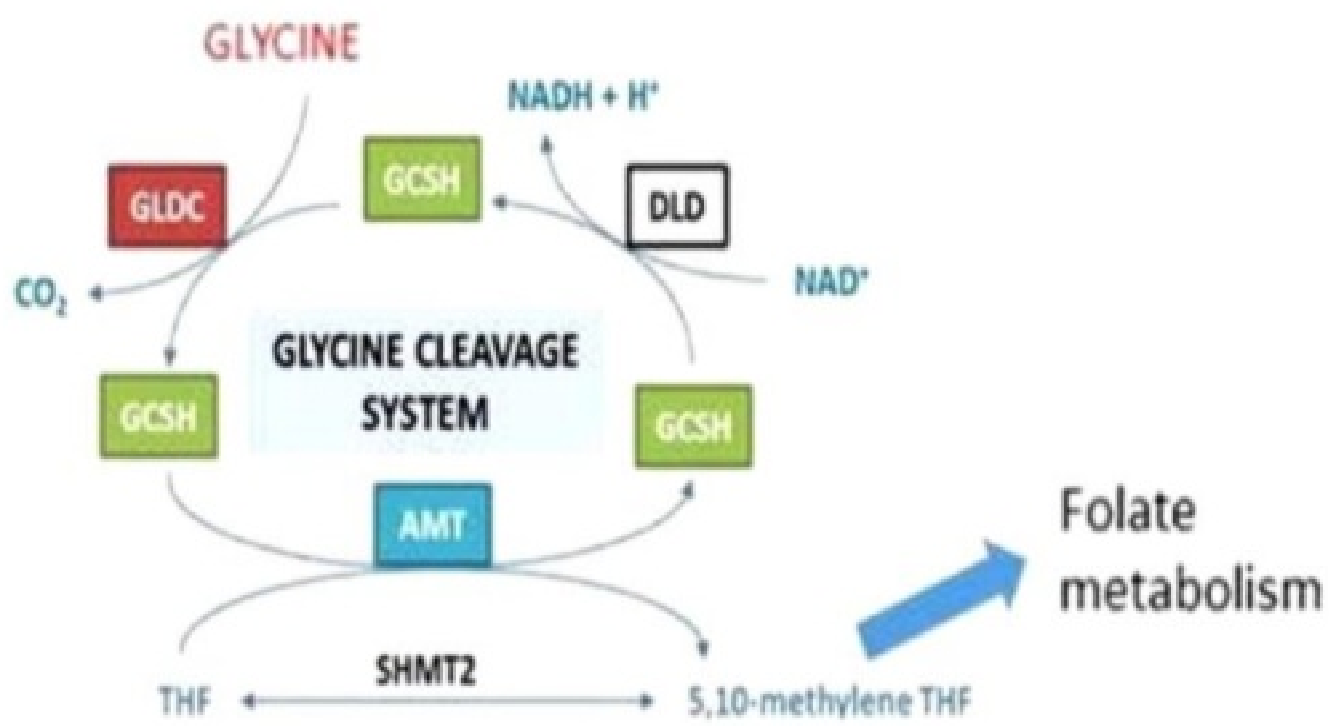
- Hornocystinuria is caused by a defect in cystathionine beta-synthase (or 13-synthase), an enzyme that participates in the metabolism of methionine, which leads to an accumulation of homocysteine.

- Symptoms include a pronounced flush of the cheeks, a tall, thin frame, lens dislocation, vascular disease, and thinning of the bones (osteoporosis).
- Intellectual disability and psychiatric disorders also may be present. Approximately 50 percent of persons with homocystinuria are responsive to treatment with vitamin Bs (pyridoxine), and these individuals tend to have a better intellectual prognosis.
- Therapy with folic acid, betaine (a medication that removes extra homocysteine from the body), aspirin, and dietary restriction of protein and methionine also may be of benefit.



4) NON-KETOTIC HYPERGLYCINEMIA:

- Non-ketotic hyperglycinemia (NKH) is a rare, genetic, metabolic disorder caused by a defect in the enzyme system that breaks down the amino acid glycine, resulting in an accumulation of glycine in the body's tissues and fluids. There is a classical form of NKH and a variant form of NKH.
- **Non-ketotic hyperglycinemia** is characterized by seizures, low muscle tone, hiccups, breath holding, and severe developmental impairment.
- It is caused by elevated levels of the neurotransmitter glycine in the central nervous system, which in turn are caused by a defect in the enzyme system responsible for cleaving the amino acid glycine.
- Drugs that block the action of glycine (e.g., dextromethorphan), a low-protein diet, and glycine-scavenging medications (e.g., sodium benzoate) may ease symptoms, but there is no cure for this severe condition.



5) MAPLE SYRUP URINE DISEASE:

- Maple syrup urine disease (MSUD) is a disorder of branched-chain amino acid metabolism that leads to the accumulation of leucine, isoleucine, valine and their corresponding oxoacids in body fluids—one result being a characteristic maple syrup smell to the urine of some patients.
- The disorder is common in the Mennonites of Pennsylvania. The classic form of MSUD presents in infancy with lethargy and progressive neurological deterioration characterized by seizures and coma.
- Treatment involves restricting proteins and feeding with formulas deficient in the branched-chain amino acids.
- Persons with MSUD may have intellectual disability despite therapy, but early and careful treatment can result in normal intellectual development.
- Milder forms of MSUD may be treated with simple protein restriction or administration of thiamin (vitamin B1).

Disorders of amino acid metabolism

Disease name	Enzyme deficient	Characteristics
Phenylketonuria (PKU)	**Classic form:** Phenylalanine hydroxylase (PAH) is deficient; **Malignant PKU:** dihydrobiopterin reductase deficiency that is needed to convert dihydro to tetrahydrobiopterin, a cofactor for PAH ; deficiency of dopamine and serotonin	Mental retardation, seizures, albinism, eczema, "mousy" odor, vomitings; Phenylketones are neurotoxic metabolites; Maternal PKU which is untreated can cause mental retardation, IUGR, microcephaly and heart defects in newborns; Patients with malignant PKU can suffer from neurological deficits even after phenylalanine restriction; Treat with dietary restriction of phenylalanine and tyrosine supplementation, Sapropterin supplementation, Pegvaliase to lyse phenylalanine
Maple syrup urine disease	Branched chain alpha ketoacid dehydrogenase; Affects the breakdown of isoleucine, leucine and valine	Poor feeding/irritability, then progressing to lethargy, intermittent apnea, cerebral edema, coma and death; "Maple syrup" odor (from accumulation of 2-hydroxyisoleucine); Elevations of valine, leucine and isoleucine, as well as l-alloisoleucine and respective ketoacids; Treatment is with protein restriction to minimum requirements, hemodialysis for acute crisis
Alkaptonuria	Homogentisate oxidase	Homogentisic acid accumulates causing pigmentation of connective tissue like sclera of eye or ear cartilage, arthritis of joints and spine, cardiac valve involvement, kidney and prostate stones. Urine turns dark/black when exposed to air; Treatment is with nitisinone, restriction of tyrosine, phenylalanine, high dose Vit C, oral biphosphonate
Homocystinuria (homocysteinemia)	Cystathionine synthase; Pyridoxine or Vit B6 deficiency	Homocysteine causes endothelial damage with thrombosis; Presents with myopia, dislocated lens, osteoporosis, developmental delay, learning disabilities, marfanoid features, megaloblastic anemia in some cases; Elevated levels of homocysteine and methionine; Treatment is with high doses of Vit B6, restriction of methionine and cysteine supplementation
Albinism	Tyrosinase	Absence of melanin pigment in melanocytes; Colorless hair, skin, lack of pigment in eye causing photophobia, increased risk of skin cancer, no neurological deficits (compare and contrast with PKU)
Tyrosinemia'	Type I or tyrosinosis is due to fumarylacetoacetate hydrolase (FAH) enzyme deficiency; Type II is due to TAT or tyrosine aminotransferase enzyme deficiency; Type III is due to HPD or 4-hydroxyphenylpyruvate dioxygenase deficiency; All three types cause decrease in the breakdown of tyrosine	Type I is most severe, beginning in infancy with failure to thrive, diarrhea, vomiting, jaundice, cabbage-like odor, recurrent nosebleeds, liver and kidney failure, rickets, hepatocellular carcinoma, peripheral neuropathy; Type II presents in early childhood with eye pain and redness, excessive tearing, photophobia, thick, painful skin on the palms of their hands and soles of their feet (palmoplantar hyperkeratosis) and intellectual disabilities; Type III presents with intellectual disability, seizures and intermittent ataxia; Diagnosis is by elevated levels of tyrosine and metabolites like phenyl lactates, phenyl pyruvates etc.; Treatment is by tyrosine restricted, low-protein diet and nitisinone